ChatGPT

Flight Planning Software for DJI Matrice 4E (Mapping & Inspection)

Overview: The DJI Matrice 4E is a high-end enterprise drone geared for geospatial mapping, surveying, and inspections. Key requirements for its mission planning software include **offline mission planning** (ability to plan/fly without internet) and **terrain following** (adjusting flight altitude to ground elevation). It should also work with the **DJI RC Plus 2** controller included with the M4E. Below we explore DJI's own tools and leading third-party solutions (favoring free or educational discounts), comparing their features, offline/terrain capabilities, controller compatibility, and pricing. A comparison table is provided at the end.

DJI Matrice 4E (right) alongside the thermal-camera Matrice 4T (left). The M4E is purpose-built for mapping with a 20MP wide camera, mechanical shutter, and RTK module for precision [1] [2] .

DJI Native Mission Planning Solutions

DJI Pilot 2 (Integrated Flight App)

Description: *DJI Pilot 2* is the default flight planning and control app on the Matrice 4 series' smart controller. It supports various automated flight route modes (waypoints, mapping grids, oblique missions, etc.) through an improved interface [3] . Pilot 2 is tightly integrated with the M4E's features (e.g. Smart Oblique Capture, 5-directional imaging).

- **Offline Planning:** Yes – Pilot 2 allows caching of maps and terrain data for offline use. If the controller is connected to the internet ahead of time, it can download **detailed offline maps and** terrain elevation data**, enabling automatic path planning even with no connection [4]. The app also features a “Local Data Mode” to fully block data transfer for secure offline operation [5].

- **Terrain Follow:** Yes – The Matrice 4 series introduced a *Real-Time Terrain Follow* feature for mapping. Pilot 2 can automatically adjust flight altitude based on terrain elevations to maintain consistent ground clearance [6] [7]. This helps ensure uniform image GSD and obstacle avoidance in mountainous areas.
- **Controller Compatibility:** Native – Pilot 2 comes pre-installed on the DJI RC Plus 2 controller included with the M4E. It is optimized for that device and requires no additional hardware.
- **Pricing:** Included – Pilot 2 is provided free with DJI enterprise drones. There is no extra cost. (Note: Internet access may be needed to download maps/terrain data initially, but no subscription required for its use.)

Notable: Pilot 2 supports the M4E’s unique capabilities like Smart 3D Capture (on-controller rough 3D models for on-site planning) [8]. It also generates in-field survey reports and highlights coverage areas to avoid gaps [9] [10]. Pilot 2 may be slightly less flexible than some PC software (e.g. limited custom waypoint counts per mission), but it’s reliable and purpose-built for DJI drones.

DJI Terra (Mapping Software) & FlightHub 2

DJI Terra is primarily a ground-based photogrammetry software, but it also offers some mission planning integration for DJI drones. Notably, a *1-year DJI Terra license is bundled with the Matrice 4E* [11] [3]. Terra allows you to plan 2D and 3D photogrammetry missions on a PC and execute them via the drone’s link, as well as to process the captured images into maps/models offline. It features high-precision camera calibration (important for the M4E’s mapping camera) and supports *offline PPK processing* of images [3]. While Terra can plan flights, it’s used mostly for data processing; you would typically still use Pilot 2 or the controller to actually fly the mission.

FlightHub 2 is DJI’s cloud-based fleet management platform. It lets you plan and synchronize missions remotely and even control drones via the cloud. FlightHub 2 has been updated with route planning functions for the Matrice 4 series [12]. However, it is an enterprise service (with limited-time free quotas for M4 purchasers) and requires internet/cloud – not ideal for offline needs. For an educational user or single-drone operation, FlightHub 2 might be overkill unless remote multi-user coordination is needed.

Summary: DJI’s native stack (Pilot 2 + Terra) covers most needs out-of-the-box: you get free, integrated mission planning with offline/terrain-follow support in Pilot 2 [7], plus Terra for advanced processing. These are highly compatible and require no additional purchase (Terra free for 1 year). The downside is limited cross-platform support (Pilot 2 only on the DJI controller) and less customization compared to some third-party solutions.

Third-Party Flight Planning Software

Several third-party apps extend the capabilities for mission planning on DJI drones. The Matrice 4E is supported by the DJI SDK, so many popular mapping apps now include M4E compatibility [13] [14]. Below are leading options:

DroneDeploy

Description: *DroneDeploy* is a widely used cloud-based drone mapping platform. It includes a mobile **DroneDeploy Flight App** for automated mission planning and data capture, with seamless upload to DroneDeploy's cloud for processing (or the images can be downloaded for other use). DroneDeploy supports a range of mission types ideal for surveying and inspection: standard 2D grids, crosshatch 3D flights, corridors, facade (vertical) scans, panoramas, etc. [15] – all of which are confirmed to work on the M4E/T [16].

- **Offline Planning:** Yes – The DroneDeploy flight app offers an *Offline Mapping* mode [17]. You can cache base maps and plan missions ahead of time; in the field, the app will function without internet. This is useful for remote areas. (The device should be logged in and the mission downloaded beforehand.)
- **Terrain Following:** Yes – DroneDeploy supports **Terrain Awareness** on the M4E [17]. This uses global elevation data to adjust the flight altitude relative to ground. Enabling terrain follow ensures consistent altitude above ground (helping maintain uniform image resolution and safe clearance in uneven terrain).
- **Controller Compatibility:** The DroneDeploy app can be installed directly on the DJI RC Plus 2 (Android) controller. The process involves downloading the APK via the controller's browser and installing it [18]. DroneDeploy provides step-by-step guidance, and once installed, you should *force-close DJI Pilot 2* and run DroneDeploy instead when flying missions [19]. (DroneDeploy specifically requires using the RC Plus 2 – it will not work via an alternate controller or iPad for the M4E [13].)
- **Pricing: Commercial** – DroneDeploy is a paid service. Plans range from individual to enterprise. However, **educational discounts are available** – verified schools/universities can apply for discounted pricing [20]. DroneDeploy also offers a free 14-day trial for new users [21]. *Expect pricing to be subscription-based*, so for long-term use an educational license could save cost.

Notable Features: DroneDeploy's strength is the end-to-end solution – the same platform can capture the data and process it into maps, 3D models, or analyses (construction progress, crop health, etc.). For example, with the M4E's RTK, DroneDeploy supports capturing RTK geo-tagged images for high accuracy maps [22]. It even has "Live Map" for quick in-field stitching of imagery [23]. The app's ease-of-use and cloud integration are excellent, though full capabilities require internet for processing and a paid account.

UgCS (Universal Ground Control Software)

Description: *UgCS* is a powerful PC-based mission planning software by SPH Engineering. It's known for handling very complex flight plans and supporting many drone platforms. UgCS runs on Windows/Mac and connects to DJI drones via a companion link (typically an Android device running "UgCS for DJI" connected to the controller, which relays commands from the PC).

- **Offline Planning:** Yes – UgCS is **fully offline capable**. The software installs locally and does not require internet for planning or flying missions [24]. You can cache maps or even use your own map data, and everything (including elevation data) is stored on your computer. This makes UgCS ideal for remote expeditions – you can plan missions "in the most remote corners of the Earth" without connectivity [24].
- **Terrain Following:** Yes – UgCS offers industry-leading terrain following features. You can import custom Digital Elevation Models (DEM/DSM) for precise terrain data [25], or use its built-in global elevation maps. UgCS lets you adjust how closely the flight follows terrain (e.g. resolution of terrain

sampling, smoothing of altitude changes) and can even handle near-vertical surfaces via a “Smart AGL” mode [26] [27]. In short, it excels at maintaining AGL altitude even in very rugged or vertical terrain – critical for safe LiDAR or photogrammetry in mountains.
- **Controller Compatibility:** UgCS supports virtually all DJI models including Matrice series. For the M4E, you would run UgCS on a laptop and use the DJI RC Plus controller connected (via USB or Wi-Fi) with the “UgCS for DJI” mobile app as a bridge. This setup is a bit more gear to manage in the field, but it allows you to execute the PC-planned mission on the drone. (UgCS cannot run on the DJI controller alone; it’s a desktop program.)
- **Pricing: Freemium/Commercial** – UgCS has multiple license tiers. **UgCS Open** is a free version with limited usage (enough to try out basic planning) [28] [29]. Paid versions (UgCS One, Pro, Enterprise) unlock advanced features and larger missions. Given that you’re an educational user, note that UgCS has an *Academic/Educational program* [30] – often universities can get discounted licenses or campus licenses. It’s worth contacting them for academic pricing. For personal or one-off use, the free tier might suffice for smaller jobs, or you can opt for a one-time license depending on project needs.

Notable Features: UgCS provides unmatched control over flight parameters. You can plan multi-battery missions seamlessly (automatic waypoints for DJI drone battery swap), define custom actions at waypoints, and use **imported KML/CSV** to create flight paths [31]. It’s great for corridor mapping, LiDAR surveys (with automatic LiDAR IMU calibration patterns), and other professional workflows [32] [33]. The trade-off is a steeper learning curve and the need for a laptop in the field. If you require maximum flexibility, offline autonomy, or planning very large areas, UgCS is a top choice.

Drone Harmony

Description: *Drone Harmony* is a specialized drone mission planning software focused on 3D flight planning for complex structures and terrain. It offers a Web planning interface and a mobile app for execution. Drone Harmony is known for its *scene-based planning* – you can define the 3D environment (terrain, buildings) and the app will generate optimized flight paths (e.g. for inspecting a cell tower, mapping a hillside, etc.).

- **Offline Planning:** *Partially.* Drone Harmony’s mobile app can execute missions in the field without internet, but mission planning typically leverages their cloud/web interface. Missions are normally designed on **Drone Harmony Web** (which requires an internet connection to use and to sync with your device). That said, you can plan in advance and then use the saved missions offline on the controller app. According to a project example, Drone Harmony allowed planning “various complicated missions for later offline use,” which was crucial for repeated missions on a remote volcano [34]. In essence, you should not count on designing new missions from scratch with no internet, but you *can* run pre-planned missions offline. (They have also introduced some offline workflow options for high-security use cases, but it may be enterprise-only.)
- **Terrain Following:** Yes – Terrain-aware planning is Drone Harmony’s flagship feature. It can automatically plan flights on a **high-resolution elevation model** that you provide or via their terrain service, rather than just low-res global data [35]. This is ideal for steep or irregular landscapes. For example, Drone Harmony can plan a single large-area mission on complex terrain using a high-res DSM, avoiding the need to split into many small grid patches [35]. It also has unique modes like *Hill Scan* to map very steep slopes or vertical faces (e.g. cliffs, dams) by keeping the camera normal to the surface [36] [37]. Users report that standard tools using 30m SRTM data were “useless” on steep volcano terrain, whereas Drone Harmony’s terrain-aware planning made it possible [38] [34].
- **Controller Compatibility:** Drone Harmony supports the DJI Smart Controllers including the RC Plus. The Drone Harmony mobile app (Android) is lightweight and can be **sideloaded on the RC Plus** – the

developers note it was designed to run smoothly on DJI's controller hardware [39] [40]. Installation involves loading the APK to the controller (similar to DroneDeploy's method). Once installed, you can use the RC's screen to run missions. (No iOS support for enterprise DJI; the app is Android-only for newer drones.)

- **Pricing: Commercial** – Drone Harmony is a subscription service. They offer different plans (with names like Basic, Professional, Enterprise) depending on number of drones and features. There isn't a publicly advertised free tier for the full product (sometimes they had a limited free plan for simple use or legacy drones). Check if they have academic pricing – you may need to contact sales. Given that Drone Harmony is feature-rich, cost might be higher than a hobby app. However, if you specifically need its advanced inspection planning or terrain tools for a project, it could be worthwhile.

Notable Features: Drone Harmony is especially powerful for **infrastructure inspections** (cell towers, bridges, building facades). It can generate waypoints that capture all angles of a structure with proper overlaps. For mapping, it shines when the area is **complex or large with elevation changes** – it eliminates data gaps and redundant overlaps by accounting for the 3D terrain in one unified plan [41] [42]. It also supports multi-mission coordination and re-flying stored missions (helpful for periodic surveys). One limitation is reliance on cloud for full functionality, which may complicate purely offline fieldwork (unless missions are pre-planned). Overall, for challenging 3D scenarios, Drone Harmony is a top-tier solution chosen by many enterprise operators for the M300/M350, and it **fully supports the Matrice 4E/4T** as well [43].

Dronelink

Description: *Dronelink* is a versatile drone automation platform known for its "scriptable" mission plans. It allows users to create highly customized flight plans using a web interface, including conditional actions, multi-part missions, and reusable components. Dronelink supports many DJI drones and offers a unified app that can run on web, mobile, and even the DJI RC.

- **Offline Planning:** *Limited.* The **Dronelink web app (planner) requires internet** to design or edit missions [44] [45]. However, you can **download missions for offline use** in the mobile app [46]. The workflow is: create or sync the mission while online, use the app's "Download for Offline" option, then you can go to the field and fly that mission without connectivity [47]. So, like Drone Harmony, ensure you prepare missions beforehand. The mobile app itself can function offline for execution as long as the mission and map data were cached.
- **Terrain Following:** Yes – Dronelink offers a *Terrain Follow* option (constant AGL altitude). It uses **Esri World Elevation** data to get ground heights [48]. When enabled, the mission waypoints are adjusted to follow the terrain. In the app you can preview the 3D flight path to see how it hugs the ground [49] [50]. **Note:** Terrain follow is a **premium feature** – according to Dronelink's docs it requires at least the "Starter" paid plan (not included in the free Hobbyist tier) [51]. Also, the elevation data won't update offline unless it was fetched when online (so always preview your mission in 3D before heading out, to ensure the elevations are loaded [52]).
- **Controller Compatibility:** Dronelink supports the DJI RC Plus 2 (Matrice 4 series controller). You can install it by downloading the APK on the controller's browser (similar to others) [53]. Notably, **iOS is not supported for the M4** [54] – you must use the Android-based controller or an Android device. Once installed, remember to close Pilot 2 when using Dronelink. The app will then connect to the drone via the controller.

- **Pricing: Freemium with add-ons.** Dronelink's model is a bit unique: they had a **free Hobbyist** option for personal, non-commercial use (recently announced as free) which allows basic missions [55]. However, support for advanced drones (like M4E, M300) and advanced features (terrain follow, etc.) often require higher-tier plans. The *"Professional"* or *"Growth"* plans are needed for the Matrice 4 series [56]. These are typically paid annually or monthly. They also sell one-time "missions" or bundles. For educational purposes, if your use is non-commercial (e.g. a student project), you might use the free tier but it may not allow M4E control – so likely a paid license is needed. Dronelink doesn't publicly list an academic discount, but it's relatively affordable compared to enterprise software (a few hundred dollars per year for pro plans).

Notable Features: Dronelink is extremely flexible: - You can chain complex actions (e.g., take a photo, then yaw 90°, take another, etc.), create conditional logic, and reuse mission blocks. This can be powerful for custom inspection routines. - It supports **multiple platform control** – e.g. plan on your desktop browser, then run on phone or the DJI controller. - The community has many shared mission templates. If you have unique mission requirements that other apps can't do out-of-the-box, Dronelink might handle it.

For mapping and surveying, Dronelink can certainly do grid and orbit missions with terrain following. It's perhaps less plug-and-play than DroneDeploy or Map Pilot, but very powerful for tech-savvy users. Ensure you test your missions in the 3D simulator preview (Dronelink offers Google Earth exports and a virtual flight view) to verify behavior before real flight.

Map Pilot Pro (by Maps Made Easy)

Description: *Map Pilot Pro* is a mobile app (iOS **and** Android) dedicated to drone mapping. It's the successor to the classic Map Pilot app, widely used by mapping professionals for its simplicity and reliability. Map Pilot Pro is designed to work closely with the Maps Made Easy processing service, but it can be used for flight planning/capture only, with images processed elsewhere if desired.

- **Offline Planning:** Yes – Map Pilot Pro was built with offline operation in mind. It offers **basemap caching** and a "Quick Map" system to preview maps offline [57]. All mission planning and flying can be done without internet once you've cached the needed map tiles. Critically, it also supports **fully offline terrain awareness** – you can download elevation data for your mission area in advance, so the app can adjust for terrain without any live connection [58] [59]. Field users report that Map Pilot is very reliable in remote locations.
- **Terrain Following:** Yes – Map Pilot Pro has an enhanced *Terrain Awareness* feature. By downloading the terrain info ahead of time (or importing your own DEM GeoTIFF), the app "will design the flight path to take changes in ground elevation into account" [60]. This means the grid flight's altitude is auto-adjusted to maintain a constant height above ground. Map Pilot even visually shows the camera footprint on the ground considering elevation – helping ensure coverage [61]. Additionally, you can import custom elevation data if higher resolution than the default SRTM is needed [62].
- **Controller Compatibility:** Map Pilot Pro supports DJI drones including the Matrice series (the developers explicitly list **Matrice 4 Series** as supported hardware [63]). The app is available on the iOS App Store and Google Play Store [64] [65]. For the RC Plus controller, you would use the Android .apk (since Google Play may not be directly accessible on DJI's enterprise RC, you can sideload the app). Alternatively, if you have a separate Android tablet with DJI SDK support, you could use that (though the Matrice 4's RC doesn't easily allow an external device control, so installing on the RC is the practical path).

- **Pricing: Freemium with service tie-in.** The app itself is free to download. You are encouraged (but not required) to use the Maps Made Easy (MME) processing service: *Map Pilot Pro is a companion app for Maps Made Easy's platform* [66] . MME offers a free processing allowance for small jobs (they mention you can process unlimited maps up to 325 images for free on MME [67]). The app will sync flight logs and missions to your MME account (FlightSync) for backup [66] . **All the flight features are included for free** – there are no separate subscription fees for offline, terrain, multi-battery, etc. (previous versions had "in-app purchase" for terrain, but Pro appears to include it). Essentially, MME hopes you'll buy processing "points" or subscriptions, but you can also just export the images to other software. For educational users, this is attractive: you get a fully functional mapping app at no cost, and if needed, MME has *very affordable edu pricing for their processing* (they often give discounts to students or have pay-as-you-go for small projects).

Notable Features: Map Pilot Pro is praised for: - **Simplicity:** Define your mapping area and desired GSD/altitude, and it takes care of camera settings, overlaps, etc., to optimize the flight. - **Multi-battery missions:** It automatically manages splitting large areas into multiple flights and will prompt for battery swap, then resume where it left off [68] [69] . - **Safety and control:** It has features like adaptive speed (slowing down if needed for camera interval), signal loss behaviors, and it captures a "terrain reference image" (ground shot) for help in processing [70] . It also supports DJI's obstruction avoidance and will warn for any issues with the planned altitude vs terrain.

Given it's free and robust, Map Pilot Pro is a top recommendation for academic use on DJI drones. The only caution is ensuring it's installed on the M4E controller. If installation is successful, it provides nearly everything needed for offline, terrain-aware mapping missions out-of-the-box.

Comparison of Key Platforms

Below is a **comparison table** summarizing the features of the discussed flight planning software for the Matrice 4E:

Software	Offline Mission Planning	Terrain Follow Support	Controller Compatibility	Pricing & Educational Notes	Notable Features
DJI Pilot 2	Yes (cache maps & data for offline use) [4] . Local Data Mode for full offline [5] .	Yes – Real-time terrain following using downloaded elevation data [7] .	DJI RC Plus 2 (comes pre-installed) [3] . Android only (enterprise controllers).	**Free** (included with drone). – No extra cost.	Integrated with all M4E features (oblique capture, AI spotting). On-controller 3D model preview and mission tweak [9] . Generates on-site reports. Easy to use; limited customization.

Software	Offline Mission Planning	Terrain Follow Support	Controller Compatibility	Pricing & Educational Notes	Notable Features
DroneDeploy	Yes – Offline maps mode supported [17] (plan and cache ahead).	Yes – Terrain Awareness uses global elevation to adjust AGL [17].	DJI RC Plus 2 (Android) via DroneDeploy Flight app [18]. *Not* on iOS for M4E [13].	**Subscription.** Free 14-day trial [21]. Edu/ nonprofit discounts available [20].	End-to-end cloud platform (flight + processing). Multiple mission types (2D, 3D, corridor, facade) [15]. Live cloud sync and "Live Map" rapid orthomosaic [23]. RTK support for high accuracy [22].
UgCS (SPH Eng.)	Yes – Fully offline desktop software [24]. No internet required in field.	Yes – Advanced terrain follow. Use custom DEMs [25]; fine control of terrain resolution & slope handling [26].	PC software (Windows/ Mac). Connect to M4E via RC (UgCS mobile link app) – supports all DJI Enterprise [71].	**Freemium/ License.** *UgCS Open* free with limited usage [29]. Paid licenses (One/Pro) for full features. Edu discounts via academic program.	Extremely powerful planning (multi-km routes, unlimited waypoints). Complex mission types (LiDAR calibration patterns, vertical scans, orbitgrams). 3D interface with mission preview. Full 3D maps and no-fly zones on map [72] [73]. Steeper learning curve; requires laptop.

Software	Offline Mission Planning	Terrain Follow Support	Controller Compatibility	Pricing & Educational Notes	Notable Features
Drone Harmony	*Partial.* Plan missions on cloud/web (requires internet); can execute pre-planned missions offline on mobile [34].	Yes – High-res terrain-aware planning. Hill-Scan mode for steep/vertical terrain [74]. Uses custom or provided elevation layers [35].	DJI RC Plus / Smart Controller (Android app can be sideloaded) [75]. Web planner on PC for design. No iOS support for M4.	**Subscription.** No free tier for enterprise drones (free trial/demo on request). Pricing tiers by features; inquire for academic use.	Best for complex 3D missions (asset inspections, mountain mapping). Scene-based automated planning eliminates data gaps [41]. Multi-mission management and repeat scheduling. Lean app runs smoothly on DJI RC [40]. Requires planning workflow with their cloud.
Dronelink	*Partial.* Missions designed online; can **download to device for offline use** [47]. No on-the-fly editing offline.	Yes – Terrain Follow (AGL) uses Esri elevation data [48]. Must configure while online; preview in 3D [49].	DJI RC Plus 2 (Android) supported via APK [53]. Not on iOS for M4E [54]. Also works on Android phones/tablets (if they could connect).	**Freemium/ Paid.** Hobbyist (free) plan exists, but **M4E requires paid Pro** plan [56]. No official edu program (non-commercial use allowed for free on supported drones).	Highly customizable "programmatic" missions (conditional logic, multi-part). Cross-platform planning with 3D simulator [49]. Strong community sharing mission templates. Good for unique use cases beyond standard grids. Some features gated by plan level (e.g. advanced camera control, terrain = paid) [51].

Software	Offline Mission Planning	Terrain Follow Support	Controller Compatibility	Pricing & Educational Notes	Notable Features
Map Pilot Pro	Yes – Designed for **fully offline** use [58]. Cache maps and even operate with no signal.	Yes – Offline terrain awareness with pre-downloaded or custom DEM [76] [60]. Adjusts flight altitude to ground automatically.	DJI RC Plus (Android) – supported via sideload or Amazon Appstore. Also on iOS (for other DJI controllers). M4E supported as listed hardware [77] [63].	**Free app.** No flight subscription required. (Tied to Maps Made Easy processing: free up to 325 images per map [78], pay-as-you-go for more; educational rates available).	Focused on mapping efficiency and safety. Multi-battery missions with auto-resume [68]. FlightSync logs and mission cloud backup. Outputs compatible with any photogrammetry software. Known for reliability in the field and easy setup. Ideal for budget-conscious mapping teams.

Table Notes: All listed software support the Matrice 4E's **default payload** (the built-in triple camera). They automatically handle camera triggering at optimal intervals. Some advanced M4E-specific camera functions (e.g. smart oblique 5-angle capture, or the built-in laser rangefinder) may not be fully accessible in third-party apps due to DJI SDK limitations [79]. For example, DroneDeploy notes that the M4E's 5-directional oblique mode and laser ranger aren't available via their app [80] – those are typically only in DJI Pilot 2. However, this doesn't hinder core mapping/inspection use, as the apps still take photos with the main camera and can use manual oblique presets if needed.

Conclusion & Recommendations

For a Matrice 4E user focused on mapping and inspection missions, there are excellent software options:

- **If you prefer to use what's included:** Start with **DJI Pilot 2**. It's already on the controller, supports offline terrain-follow missions [7], and is tailored to the M4E's capabilities. You won't incur extra cost. Pilot 2 is quite capable for standard grid mapping and waypoint flights. Use the free **DJI Terra license** to process your maps/models if you don't have other software [3].

- **For advanced academic mapping on a budget: Map Pilot Pro** is highly recommended. It gives you offline, terrain-aware flight planning with no software cost, and the option of using Maps Made Easy's cloud processing (which has generous free allowances and edu discounts). Many university programs favor this app for teaching drone mapping due to its simplicity and cost-effectiveness.

- **For comprehensive professional projects or research:** consider **UgCS** or **Drone Harmony** depending on your needs:

- Choose **UgCS** for full offline autonomy, huge or complex area missions, or if you need to integrate custom terrain data extensively. It's a great choice for research projects in remote areas with challenging terrain.
- Choose **Drone Harmony** if your work involves very complex terrain or structures (like the volcano mapping example) and you need sophisticated mission planning logic to ensure complete coverage. It's excellent for repeatable inspection routines and can tackle terrain where other tools struggle [34] [74] .

- **If an integrated cloud solution fits your needs and budget: DroneDeploy** is a strong option, especially with an educational discount. It simplifies the workflow (planning to processing) and supports the M4E well [16] . The trade-off is cost, but for an organization or lab, the efficiency might justify it.

- **For custom or experimental flight automation: Dronelink** is unique. It might appeal to those who want to tinker with mission logic or integrate the drone into custom applications. If your project involves writing custom survey patterns or integrating with other data, Dronelink's flexible API-like approach is powerful.

In summary, the Matrice 4E is supported by a rich ecosystem of flight planning tools. DJI's own software ensures you have baseline offline and terrain-follow capabilities [7] . Third-party apps can enhance productivity, especially for specialized use cases or if you have constrained funding (some free tools are available). Consider the complexity of your missions, whether you need cloud processing, and your budget:

- *For straightforward mapping with minimal fuss:* Pilot 2 or Map Pilot Pro (free, offline).
- *For enterprise-grade workflows and processing:* DroneDeploy (with edu discount) or DJI Terra (included).
- *For complex terrain and unlimited offline control:* UgCS or Drone Harmony (powerful but higher learning curve/cost).
- *For custom automation:* Dronelink (requires some investment and learning, but very flexible).

By comparing the options and possibly testing their trials/free versions, you can identify which platform best meets your needs for the Matrice 4E. All the above options will handle mapping, surveying, and inspections – the differences are in the fine features and user experience. Good luck with your missions!

Sources:

- DJI Enterprise – *Matrice 4 Series Features & Software* [4] [3] [6]
- DroneDeploy Support – *Matrice 4E/T Compatibility* [17] [13] ; Pricing/Education FAQ [20]
- SPH Engineering – *UgCS Product Page* (offline and terrain features) [24] [25]
- Drone Harmony Blog – *Terrain Aware Planning Use-Case* [34] [74]
- Dronelink Support – *Terrain Follow and Offline Use* [48] [47]
- Maps Made Easy – *Map Pilot Pro Features* [58] [60]
- DJI Documentation – *GS Pro (Ground Station) and Pilot apps* [7] [81] (verification of supported drones)

- Additional references from DSLRPros and Dronefly – *Flight Planning Software Guides* [82] [83]. (General context on software capabilities)

---

[1] [4] [5] [7] [8] DJI Matrice 4 Series Brings Intelligence to Aerial Operations - DJI Enterprise
https://enterprise.dji.com/news/detail/matrice-4-series-release

[2] [13] [15] [16] [17] [18] [19] [22] [23] [79] [80] Matrice 4 Enterprise/Matrice 4 Thermal (M4E/T) – DroneDeploy
https://help.dronedeploy.com/hc/en-us/articles/30917880312343-Matrice-4-Enterprise-Matrice-4-Thermal-M4E-T

[3] [9] [10] [11] [12] Top Features of the Matrice 4 Series
https://enterprise-insights.dji.com/blog/top-features-of-the-matrice-4-series

[6] Urban Planning- Surveying & Mapping - DJI Enterprise
https://enterprise.dji.com/geospatial/urban-planning

[14] [53] [54] [56] Matrice 4 Series (M4E, M4T) Support Overview – Dronelink
https://support.dronelink.com/hc/en-us/articles/41394743896467-Matrice-4-Series-M4E-M4T-Support-Overview

[20] [21] DroneDeploy Pricing: Reality Capture Platform Plans for Construction, Energy & Agriculture
https://www.dronedeploy.com/pricing

[24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [71] [72] [73] UgCS - Drone flight planning software
https://www.sphengineering.com/flight-planning/ugcs

[34] [36] [37] [38] [74] Volcano Disaster Prevention: Drone Harmony's Terrain Workflow is a Major Help in Japan
https://www.droneharmony.com/post/volcano-disaster-prevention

[35] [39] [40] [41] [42] [75] Why you should be using Drone Harmony with your DJI M300
https://www.droneharmony.com/post/dji-m300

[43] Supported Hardware | Drone Harmony
https://www.droneharmony.com/supported-hardware

[44] FAQ - Dronelink
https://support.dronelink.com/hc/en-us/sections/360004427714-FAQ

[45] offline use confusion - Dronelink
https://support.dronelink.com/hc/en-us/community/posts/9933814167059-offline-use-confusion

[46] [47] Download Missions for Offline Use in Dronelink Mobile App
https://support.dronelink.com/hc/en-us/articles/4406012655763-Offline-Support-Download-Missions-for-Offline-Use-in-Dronelink-Mobile-App

[48] [49] [50] [51] [52] Terrain Follow (AGL) in Map Missions – Dronelink
https://support.dronelink.com/hc/en-us/articles/6896503977363-Terrain-Follow-AGL-in-Map-Missions

[55] I almost quit the hobby because I was done with Dronelink's gated ...
https://www.reddit.com/r/UAVmapping/comments/1mxcp01/i_almost_quit_the_hobby_because_i_was_done_with/

[57] [58] [59] [61] [62] [63] [64] [65] [66] [67] [68] [69] [70] [76] [77] [78] Map Pilot Pro
https://dronesmadeeasy.com/map-pilot?srsltid=AfmBOooMzG9vA0IK-vvLJlhKsYwi9YNRNMXN23Dk3nyFwYfvLFOglfPQ

[60] Terrain Awareness - Drones Made Easy
https://support.dronesmadeeasy.com/hc/en-us/articles/211810943-Terrain-Awareness

81 GS Pro - Supported Products - DJI
https://www.dji.com/ground-station-pro/supported-product

82 What is the Difference Between Pix4D, Drone Deploy, and DJI Terra?
https://www.dslrpros.com/blogs/drone-trends/what-is-the-difference-between-pix4d-drone-deploy-and-dji-terra?srsltid=AfmBOooiC0ZcSkRoMr1Hoeq1REpgqrpCJp5dwKWz4F1Z88KAM0ol29HO

83 Understanding Drone Flight Path Planning Software - Dronefly
https://www.dronefly.com/blogs/news/understanding-drone-flight-path-planning-software?srsltid=AfmBOoo6IY2XmVKC_AtTyl9XTahZO1-u-jeaRXHA-cd03z3tNPoxePAW